Raptor 2

How does a rocket engine work?

Two liquids colder than −160 °C are pumped up to 300 bar. A small share of each burns in its own preburner, and those hot gases spin the pumps. Then both gases meet in the main chamber, and the bell turns their pressure into 230 tonnes of thrust.

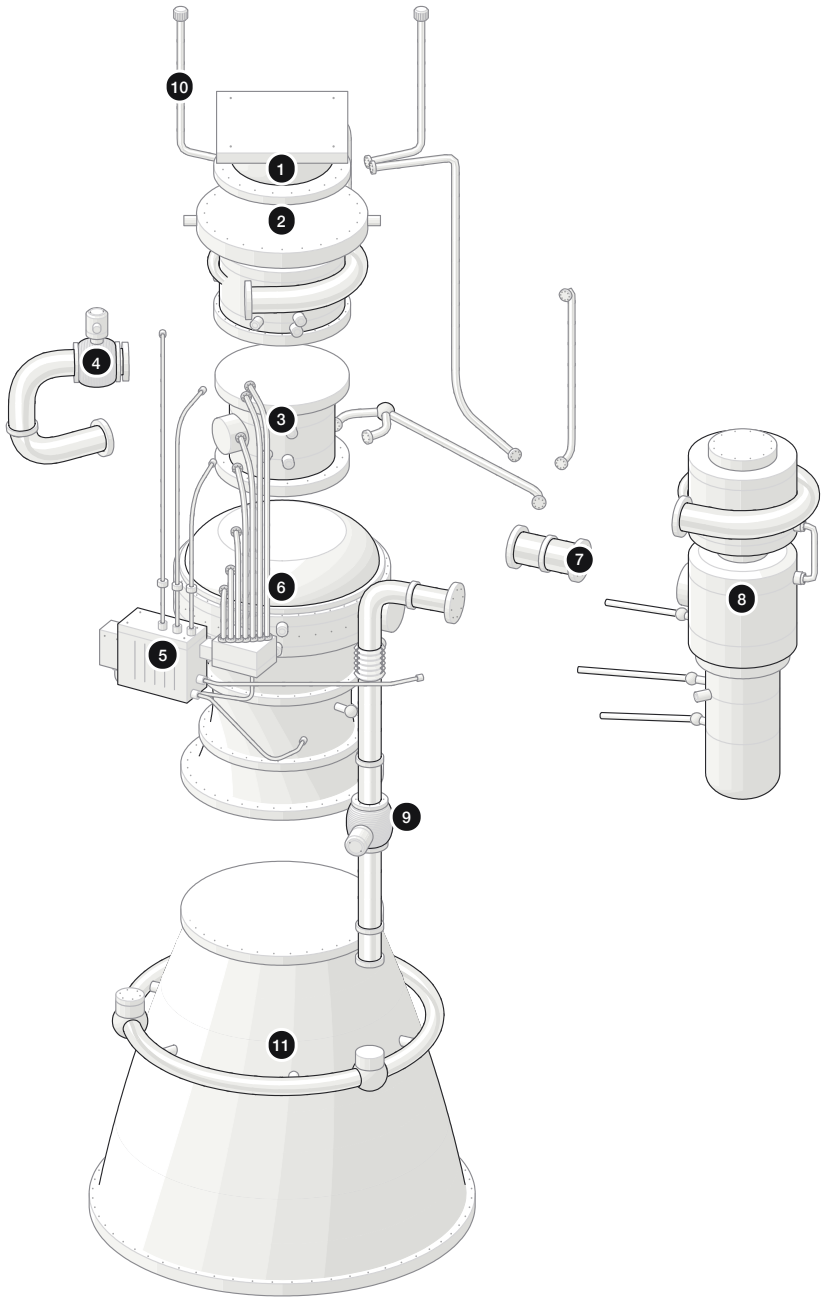
230 tf
THRUST AT SEA LEVEL

300 bar
MAIN CHAMBER PRESSURE

3.6 : 1
OXYGEN TO METHANE, BY MASS

1,630 kg
DRY MASS

1.3 m
NOZZLE EXIT



01 **Gimbal and thrust puck**
Hands the thrust to the stand and lets the engine steer.

02 **Oxygen turbopump**
Raises the liquid oxygen to chamber pressure and beyond.

03 **Oxygen-rich preburner**
Burns a trace of methane in all the oxygen to drive the oxygen pump.

04 **Oxygen duct and main valve**
Carries the pumped oxygen to its preburner; the valve meters it.

05 **Engine controller and harness**
Times the valves and igniters and reads the pump speed.

06 **Injector and main chamber**
Where the two hot gases meet and burn at 300 bar.

07 **Hot-gas manifold**
Takes the fuel-rich turbine exhaust into the injector.

08 **Methane turbopump and fuel-rich preburner**
Raises the methane, driven by burning a little oxygen in all of it.

09 **Downcomer and main fuel valve**
Sends pumped methane down to cool the nozzle first.

10 **Plumbing**
Sense, purge, igniter and pressurisation lines between the parts.

11 **Nozzle**
A methane-cooled bell that turns pressure into speed.